

Entropy and the Geometry of Chaos

From Shannon's coin flips to Pesin's entropy formula

Lecture notes, two hours

Abstract

Let $T(x) = 2x$ on $\mathbb{R}/\mathbb{Z}$. Knowing x to n binary digits determines Tx to only $n - 1$, so T destroys one bit per iterate: an entropy of $\log 2$. Separately, T expands lengths by a factor of 2, an expansion rate of $\log |T'| = \log 2$. Two entirely unrelated computations returned the same number, and this is no accident.

These notes explain why. We define metric entropy, the standard gauge of how chaotic a system is, establish its basic properties, and see via the Shannon–McMillan–Breiman and Brin–Katok theorems that it can be read off at a single typical orbit. We then introduce Lyapunov exponents, which record the average rates at which a differentiable map expands and contracts, and prove Ruelle's inequality: information is created no faster than space is stretched. Pesin's entropy formula says that for a smooth system preserving a smooth measure the inequality is an equality, and we explain both why that is true and what goes wrong without the smoothness of the measure.

Prerequisites: linear algebra, basic probability, and comfort with limits. Familiarity with measure theory is helpful but not assumed.

HOW THESE NOTES ARE ORGANIZED. The material is split into two hours, with the break after §5. Timing estimates appear flush right under each section heading; they sum to about 119 minutes, so there is very little slack. Marked with ► are the handful of computations worth doing live on the board rather than reading out. Section 9 collects three asides that were cut for time and can be restored if a section runs short.

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Lecture 1: What entropy is

1 The puzzle

≈ 5 minutes

Take a number $x \in [0, 1)$, known to fifty binary digits, and apply the map

$$T : \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{R}/\mathbb{Z}, \quad T(x) = 2x.$$

In binary, T is the shift: if $x = 0.b_1b_2b_3\dots$ then $Tx = 0.b_2b_3b_4\dots$. So knowing x to fifty digits tells you Tx to forty-nine, T^2x to forty-eight, and after fifty steps you know nothing at all. The map destroys exactly one binary digit — one *bit* — per iterate. Call this an information loss rate of $\log 2$.

Now differentiate. We have $|T'| \equiv 2$, so an interval of length δ is carried to an interval of length 2δ , and after n steps to one of length $2^n\delta$. The map expands lengths at exponential rate $\log |T'| = \log 2$.

These two computations have nothing to do with each other. One counts symbols in a base-2 expansion; the other takes a derivative. They returned the same number.

► *On the board:* write $\log 2 = \log 2$ on the board, label the left side “information” and the right side “expansion”, and leave it there for the rest of both lectures.

The goal of these two hours is to explain that coincidence: to define what the left-hand side means for a general dynamical system, to define what the right-hand side means, and then to prove a theorem (Pesin’s) saying that under the right hypotheses they agree — together with a theorem (Ruelle’s) saying that in general the left side is at most the right, and examples showing how badly the equality can fail.

Plan

Lecture 1 — what entropy is. Shannon’s entropy of a random variable, with several different explanations of the same quantity; the Kolmogorov–Sinai entropy of a dynamical system; and the Shannon–McMillan–Breiman and Brin–Katok theorems, which say that entropy is not merely an average but is visible along a single typical orbit.

Lecture 2 — where entropy comes from. Lyapunov exponents and Oseledets’ theorem; Ruelle’s inequality with a complete proof; Pesin’s entropy formula, its proof idea, and the examples where it fails.

Throughout there is one running example, $T(x) = 2x$ on $\mathbb{R}/\mathbb{Z}$, and one recurring slogan:

Entropy is the exponential growth rate of the number of things that actually happen.

2 Shannon entropy

≈ 22 minutes

2.1 What should “information” mean?

You are about to observe a random variable X taking values in a finite set $\{1, \dots, k\}$ with probabilities $p_1, \dots, p_k$. How much do you learn, on average, when you see the outcome?

Before writing a formula, ask what properties the answer $H(p_1, \dots, p_k)$ ought to have. It should be continuous in p ; it should be largest, for each fixed k , when all outcomes are equally likely; appending an impossible outcome should change nothing; and learning the pair (X, Y) should cost the same as learning X and then learning Y given X . That last requirement is the substantive one.

Theorem 2.1 (Shannon 1948; Khinchin 1953). *If H satisfies the four requirements above, then there is a constant $c > 0$ with*

$$H(p_1, \dots, p_k) = -c \sum_{i=1}^k p_i \log p_i.$$

The constant merely chooses units. Taking $\log_2$ gives *bits*; taking $\log_e$ gives *nats*. We will use natural logarithms throughout, with the convention $0 \log 0 = 0$, and translate into bits when it helps intuition.

The point worth stressing to the audience is that we did not choose this formula for convenience. Four mild and independently reasonable demands force it.

2.2 The definition, and first examples

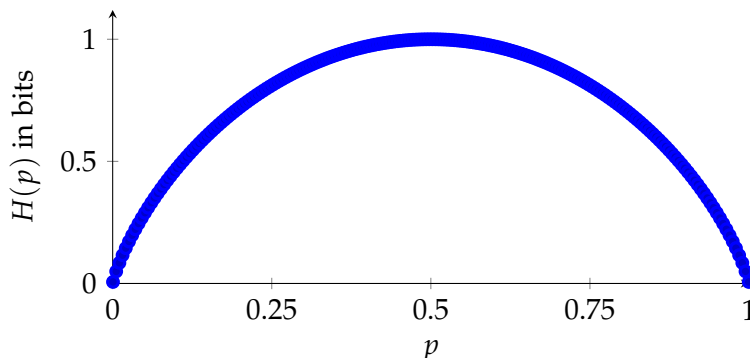
Definition 2.2. The **Shannon entropy** of a probability vector $p = (p_1, \dots, p_k)$ is

$$H(p) = - \sum_{i=1}^k p_i \log p_i = \sum_i p_i \log \frac{1}{p_i}.$$

Immediate computations: a deterministic outcome has $H = 0$; a fair coin has $H = \log 2$, one bit; a fair die has $H = \log 6 \approx 2.58$ bits; the uniform distribution on k symbols has $H = \log k$, and this is the maximum. For a biased coin,

$$H(p) = -p \log p - (1 - p) \log(1 - p),$$

whose graph is the following.



The graph is concave, symmetric about $p = \frac{1}{2}$, and vanishes only at the endpoints: certainty carries no information. Worth pointing at: the slope is infinite at $p = 0$ and $p = 1$. A nearly-certain coin surprises you only rarely, but when it does, it surprises you enormously, and the product of a small probability with a large surprise does not vanish as fast as one might guess.

2.3 Interpretation 1: average surprise

Define the *surprise* of an event of probability p to be $\log \frac{1}{p}$. Why that function? Surprise ought to decrease in p , ought to vanish when $p = 1$, and ought to be additive on independent events, since $\log \frac{1}{pq} = \log \frac{1}{p} + \log \frac{1}{q}$. Up to a multiplicative constant, $\log \frac{1}{p}$ is the only continuous function with those three properties. Then

$$H(X) = \mathbb{E} \left[\log \frac{1}{p(X)} \right]$$

is precisely the average surprise. This is the shortest gloss on the formula, and the least useful — it explains the shape but not why anyone should care.

2.4 Interpretation 2: twenty questions, and coding

Here is a version with operational content. I pick a random X ; you identify it by asking yes/no questions. How many do you need on average?

Take $X \in \{a, b, c, d\}$ with probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}$, and the code

$$a \mapsto 0, \quad b \mapsto 10, \quad c \mapsto 110, \quad d \mapsto 111.$$

No codeword is a prefix of another, so the code is uniquely decodable, and the expected length is

$$\frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{8}(3) = 1.75 \text{ bits} = H(X).$$

Theorem 2.3 (Source coding theorem). *Every prefix-free binary code for X has expected length at least $H(X)$ bits, and some prefix-free code has expected length less than $H(X) + 1$.*

So $H(X)$ is the true cost, in bits, of describing X — not a convention but a hard limit. The heuristic behind the theorem is that an outcome of probability p deserves a codeword of length $\log_2 \frac{1}{p}$: there are 2^ℓ words of length ℓ , so if we want to reserve a p -fraction of them we need $\ell = \log_2 \frac{1}{p}$. Averaging gives H .

2.5 Interpretation 3: counting typical sequences

This is the interpretation that generalizes to everything else in these notes, so it deserves the most time.

Draw $X_1, \dots, X_n$ independently from p . By the law of large numbers a typical sequence contains symbol i about np_i times. Ask how many such sequences there are:

$$\#\{\text{sequences with exactly } np_i \text{ copies of } i\} = \binom{n}{np_1, \dots, np_k} = \frac{n!}{\prod_i (np_i)!}.$$

Apply Stirling in the form $\log m! = m \log m - m + O(\log m)$. The linear terms cancel because $\sum_i np_i = n$, and

$$\log \frac{n!}{\prod_i (np_i)!} = n \log n - \sum_i np_i \log(np_i) + O(\log n) = -n \sum_i p_i \log p_i + O(\log n).$$

► *On the board:* do this computation. It is three lines, entirely elementary, and it is the moment the whole subject becomes visible. Everything afterwards is this calculation in disguise.

We have proved:

$$\#\{\text{typical sequences of length } n\} = e^{nH(p)+o(n)}.$$

2.6 The asymptotic equipartition property

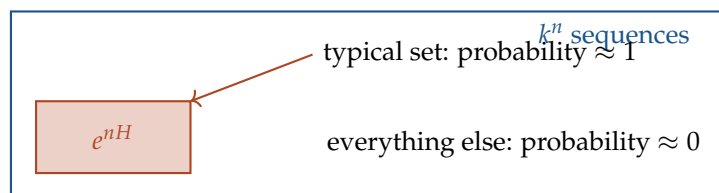
The counting statement upgrades to a probabilistic one.

Theorem 2.4 (AEP). *For i.i.d. $X_1, X_2, \dots$ and any $\delta > 0$ there are sets $A_n \subset \{1, \dots, k\}^n$ with*

$$\mathbb{P}(A_n) \rightarrow 1, \quad |A_n| = e^{n(H \pm \delta)}, \quad \mathbb{P}(x) = e^{-n(H \pm \delta)} \text{ for } x \in A_n.$$

In words: out of k^n possible sequences only about e^{nH} ever actually occur, and those are essentially equally likely. The proof is the law of large numbers applied to $-\frac{1}{n} \sum_i \log p(X_i)$.

A concrete instance to quote: flip a coin with $p = 0.1$ one thousand times. There are 2^{1000} possible outcomes. Since $H(0.1) \approx 0.469$ bits, only about 2^{469} of them are typical. The typical set is a 2^{-531} fraction of everything — an unimaginably small proportion — and yet it carries essentially all of the probability.



That gap between 2^{1000} and 2^{469} is exactly what compression exploits, and it is the sentence to leave the audience with:

Entropy is the exponential growth rate of the number of things that actually happen.

Every subsequent notion of entropy in these notes is this sentence with a different meaning assigned to “things that happen”.

Remark 2.5 (Interpretation 4). Boltzmann’s tombstone reads $S = k_B \log W$, where W counts the microstates consistent with an observed macrostate. That is the display above, with “typical set” replaced by “microstates” — found seventy years before Shannon, in a different subject, and for different reasons.

2.7 Joint and conditional entropy

For a pair (X, Y) set $H(X, Y) = -\sum_{x,y} p(x, y) \log p(x, y)$ and

$$H(Y | X) = \sum_x p(x) H(Y | X = x).$$

Three facts, all proved by convexity of $t \log t$ or by direct manipulation:

chain rule: $H(X, Y) = H(X) + H(Y | X),$
subadditivity: $H(X, Y) \leq H(X) + H(Y),$ with equality iff $X \perp Y,$
conditioning reduces entropy: $H(Y | X) \leq H(Y).$

Together with one further, entirely crude bound, these are all we shall need:

$$H(X) \leq \log \#\{\text{values } X \text{ takes with positive probability}\}. \quad (1)$$

That is just the statement that the uniform distribution maximizes entropy. It looks too weak to prove anything; it is in fact the engine of Ruelle's inequality, and the place where all of the slack in that inequality lives.

2.8 An example with real numbers: English

Treat English text as a stationary process on 27 symbols, the letters and a space.

model	entropy (bits per character)
uniform on 27 symbols	$\log_2 27 \approx 4.75$
single-letter frequencies	≈ 4.07
digram frequencies	≈ 3.36
trigram frequencies	≈ 2.77
Shannon's human-prediction experiment (1951)	$\approx 1.0\text{--}1.3$

Each row conditions on more of the past and the entropy drops, which is “conditioning reduces entropy” in the wild. The limiting value of roughly 1.1 bits per character says that the number of *plausible* English texts of length n grows like $2^{1.1n}$ rather than 27^n . This is why text compresses by about seventy-five percent, and why crossword puzzles have unique solutions.

3 Kolmogorov–Sinai entropy

≈ 20 minutes

3.1 Kolmogorov's problem

A *measure-preserving system* is a triple (X, μ, T) with μ a probability measure and $T : X \rightarrow X$ satisfying $\mu(T^{-1}A) = \mu(A)$ for every measurable A . Two systems are *isomorphic* if there is a measure-preserving bijection, defined off a null set, intertwining them.

Question (Kolmogorov, 1958). Is the fair-coin shift $B(\frac{1}{2}, \frac{1}{2})$ isomorphic to the biased shift $B(\frac{1}{3}, \frac{2}{3})$?

Both are mixing, both have countable Lebesgue spectrum, and von Neumann's spectral invariants — the only general tools available at the time — cannot distinguish them. Kolmogorov's insight was to import Shannon's entropy into ergodic theory and produce an invariant of a completely different character. The two shifts turn out to have entropies $\log 2 \approx 0.693$ and $H(\frac{1}{3}) \approx 0.637$, so they are not isomorphic.

It is worth telling the audience this story before the definition. The definition is otherwise unmotivated, and this is a case where the history is the motivation.

3.2 Partitions are measuring devices

One cannot observe a point $x \in X$ exactly. Instead there is an instrument of finite precision: a finite partition

$$\mathcal{P} = \{P_1, \dots, P_k\}, \quad X = P_1 \sqcup \dots \sqcup P_k,$$

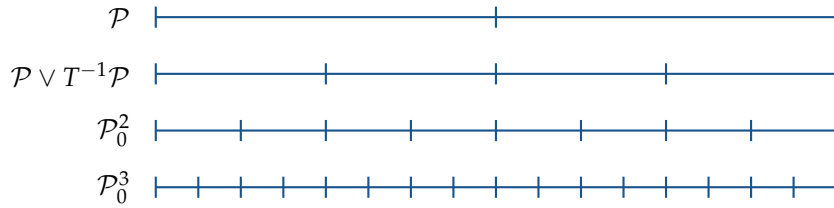
and observing x reveals only which cell contains it. The information gained from one such measurement is $H_\mu(\mathcal{P}) = -\sum_i \mu(P_i) \log \mu(P_i)$, the Shannon entropy of the vector $(\mu(P_i))_i$.

Now run the dynamics and measure at times $0, 1, \dots, n-1$. Knowing the whole *itinerary* amounts to knowing which cell of

$$\mathcal{P}_0^{n-1} := \mathcal{P} \vee T^{-1}\mathcal{P} \vee \dots \vee T^{-(n-1)}\mathcal{P} = \{P_{i_0} \cap T^{-1}P_{i_1} \cap \dots \cap T^{-(n-1)}P_{i_{n-1}}\}$$

contains x . This common refinement, or *join*, is the central construction.

For the doubling map with $\mathcal{P} = \{[0, \frac{1}{2}), [\frac{1}{2}, 1)\}$ the refinements are transparent:



The atoms of $\mathcal{P}_0^{n-1}$ are exactly the dyadic intervals $[j2^{-n}, (j+1)2^{-n})$ — that is, knowing the first n binary digits of x . There are 2^n of them, each of measure 2^{-n} , so

$$H_\mu(\mathcal{P}_0^{n-1}) = n \log 2. \quad (2)$$

3.3 The definition

Proposition 3.1. *The sequence $a_n := H_\mu(\mathcal{P}_0^{n-1})$ is subadditive, $a_{m+n} \leq a_m + a_n$. Consequently $\lim_n a_n/n$ exists and equals $\inf_n a_n/n$.*

Reason. $\mathcal{P}_0^{m+n-1} = \mathcal{P}_0^{m-1} \vee T^{-m}\mathcal{P}_0^{n-1}$, so subadditivity of H gives $a_{m+n} \leq a_m + H_\mu(T^{-m}\mathcal{P}_0^{n-1})$, and the second term equals a_n because μ is T -invariant. Fekete's lemma supplies the limit. $\square$

Definition 3.2. The entropy of T relative to $\mathcal{P}$, and the **Kolmogorov–Sinai entropy** of T , are

$$h_\mu(T, \mathcal{P}) = \lim_{n \rightarrow \infty} \frac{1}{n} H_\mu(\mathcal{P}_0^{n-1}), \quad h_\mu(T) = \sup_{\mathcal{P} \text{ finite}} h_\mu(T, \mathcal{P}).$$

Three ways to read the same formula, which should all be said out loud:

- (i) the number of bits per unit time extracted by the instrument $\mathcal{P}$, then optimized over instruments;
- (ii) the exponential growth rate of the number of *distinguishable* orbit segments of length n ;
- (iii) by the AEP, the number of itineraries that actually occur is about e^{nh} .

Reading (iii) is the slogan from §2 with “things that happen” meaning “length- n itineraries”.

3.4 Example: the doubling map

By (2), $h_\mu(T, \mathcal{P}) = \log 2$ for the two-interval partition. But the definition demands a supremum over all finite partitions, and a priori some cleverer instrument might do better.

Theorem 3.3 (Kolmogorov–Sinai generator theorem). *If $\mathcal{P}$ is a generator — meaning that $\bigvee_{n \geq 0} T^{-n} \mathcal{P}$ generates the full σ -algebra modulo null sets — then $h_\mu(T) = h_\mu(T, \mathcal{P})$.*

Here the atoms of $\mathcal{P}_0^{n-1}$ are intervals of length $2^{-n} \rightarrow 0$, so $\mathcal{P}$ generates and

$$h_\mu(T) = \log 2 = \log |T'|.$$

Without this theorem entropy would be uncomputable in practice; with it, a single good instrument suffices. It is the reason the subject has examples at all.

3.5 Example: Bernoulli and Markov shifts

Let $X = \{1, \dots, k\}^{\mathbb{N}}$ with the shift σ and the product measure $\mu = p^{\otimes \mathbb{N}}$, and let $\mathcal{P}$ be the partition according to the 0th coordinate. Then $\mathcal{P}_0^{n-1}$ partitions by the first n coordinates, and its atoms are cylinders of measure $p_{i_0} \cdots p_{i_{n-1}}$. Independence gives $H_\mu(\mathcal{P}_0^{n-1}) = nH(p)$ exactly, so

$$h_\mu(\sigma) = H(p) = - \sum_i p_i \log p_i.$$

Kolmogorov–Sinai entropy therefore generalizes Shannon entropy, and Kolmogorov’s original question is answered.

For a stationary Markov chain with transition matrix (p_{ij}) and stationary vector π , the same computation with dependence gives

$$h_\mu(\sigma) = - \sum_{i,j} \pi_i p_{ij} \log p_{ij},$$

the average over states of the entropy of the next symbol given the present one — once again “conditioning reduces entropy”, and once again strictly less than the i.i.d. value with the same marginal.

3.6 Example: rotations have entropy zero

Let $R_\alpha(x) = x + \alpha$ on $\mathbb{R}/\mathbb{Z}$ with α irrational, and let $\mathcal{P}$ be a partition into k intervals. Each $R_\alpha^{-i} \mathcal{P}$ is again a partition into k intervals, determined by k endpoints, so $\mathcal{P}_0^{n-1}$ is cut out by at most kn points and hence has at most kn atoms. By (1),

$$H_\mu(\mathcal{P}_0^{n-1}) \leq \log(kn) \quad \implies \quad h_\mu(R_\alpha, \mathcal{P}) \leq \lim_n \frac{\log(kn)}{n} = 0.$$

Interval partitions generate, so $h_\mu(R_\alpha) = 0$.

This example is worth dwelling on, because it supplies the geometric meaning of entropy. The number of distinguishable itineraries grows *linearly*, not exponentially. Nearby points stay nearby forever; the rotation never separates them. Positive entropy is precisely exponential separation of orbits, and that is the bridge to Lecture 2.

3.7 What entropy is good for

- *Invariance.* Isomorphic systems have equal entropy. This was the whole point.
- *Powers.* $h_\mu(T^n) = n h_\mu(T)$ for $n \geq 1$, and $h_\mu(T^{-1}) = h_\mu(T)$ when T is invertible. Both will be used in the proof of Ruelle's inequality.
- *Ornstein's theorem (1970).* Among Bernoulli shifts, entropy is a *complete* invariant: equal entropy implies isomorphic. So the uniform shift on four symbols is isomorphic to $B(\frac{1}{2}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8})$ on five, both having entropy $2 \log 2$; one really can be recoded into the other.
- *Variational principle.* For a continuous map of a compact space, $h_{\text{top}}(T) = \sup_\mu h_\mu(T)$ over invariant measures: topological complexity is the supremum of the statistical complexities.

4 Entropy at a single point

≈ 11 minutes

4.1 Shannon–McMillan–Breiman

Everything so far has been an average: $H_\mu(\mathcal{P}_0^{n-1})$ is an integral over all of X . But the AEP was a statement about *one* sequence, not an average. Write $\mathcal{P}^n(x)$ for the atom of $\mathcal{P}_0^{n-1}$ containing x — the set of points indistinguishable from x by n measurements. Is it true that $\mu(\mathcal{P}^n(x)) \approx e^{-nh}$?

Theorem 4.1 (Shannon 1948; McMillan 1953; Breiman 1957). *Let (X, μ, T) be ergodic and $\mathcal{P}$ a finite partition. Then for μ -a.e. x , and also in $L^1(\mu)$,*

$$\lim_{n \rightarrow \infty} -\frac{1}{n} \log \mu(\mathcal{P}^n(x)) = h_\mu(T, \mathcal{P}).$$

The left-hand side is the information content per unit time of the itinerary of x . The theorem says that this random quantity is almost surely constant, and that the constant is the entropy.

Two sanity checks, both worth doing aloud. For the doubling map $\mathcal{P}^n(x)$ is the dyadic interval of length 2^{-n} , so $-\frac{1}{n} \log \mu(\mathcal{P}^n(x)) = \log 2$ exactly, for every x and every n . For a Bernoulli shift,

$$-\frac{1}{n} \log \mu(\mathcal{P}^n(x)) = -\frac{1}{n} \sum_{i < n} \log p_{x_i} \longrightarrow -\sum_i p_i \log p_i$$

by the law of large numbers. So SMB is a genuine generalization of the law of large numbers to dependent processes. (A sketch of the proof is in §9 if there is time.)

4.2 Bowen balls

Partitions are awkward: the answer depends on $\mathcal{P}$, and the cell boundaries are arbitrary. On a metric space one can do better.

Definition 4.2. The **Bowen ball** of order n and radius ε at x is

$$B_n(x, \varepsilon) = \{y : d(T^i x, T^i y) < \varepsilon \text{ for } 0 \leq i < n\},$$

equivalently the ball of radius ε about x in the metric $d_n(x, y) = \max_{0 \leq i < n} d(T^i x, T^i y)$.

This is the set of points whose orbits *shadow* that of x for n steps at resolution ε : the points one cannot distinguish from x using n observations of accuracy ε . It is a partition-free notion of “same itinerary”.

Compute it in the two extreme examples. For the doubling map, $d(T^i x, T^i y) = 2^i d(x, y)$ as long as the points remain close, so the binding constraint is $i = n - 1$ and

$$B_n(x, \varepsilon) = (x - \varepsilon 2^{-(n-1)}, x + \varepsilon 2^{-(n-1)}), \quad \mu(B_n(x, \varepsilon)) = 2\varepsilon e^{-(n-1)\log 2}.$$

Hence $-\frac{1}{n} \log \mu(B_n(x, \varepsilon)) \rightarrow \log 2 = h_\mu(T)$: the Bowen ball shrinks at exactly the entropy rate. For an irrational rotation, $d(R^i x, R^i y) = d(x, y)$ for every i , so $B_n(x, \varepsilon) = B(x, \varepsilon)$ for every n — it does not shrink at all — and $-\frac{1}{n} \log \mu(B_n) \rightarrow 0 = h_\mu(R_\alpha)$.

In both cases the decay rate of $\mu(B_n)$ recovered the entropy. That is a theorem.

4.3 Brin–Katok

Theorem 4.3 (Brin–Katok 1983). *Let T be a continuous map of a compact metric space and μ an ergodic invariant measure. Then for μ -a.e. x ,*

$$\lim_{\varepsilon \rightarrow 0} \limsup_{n \rightarrow \infty} -\frac{1}{n} \log \mu(B_n(x, \varepsilon)) = \lim_{\varepsilon \rightarrow 0} \liminf_{n \rightarrow \infty} -\frac{1}{n} \log \mu(B_n(x, \varepsilon)) = h_\mu(T).$$

Place the two pointwise theorems side by side:

$$\begin{array}{ll} \text{SMB:} & \mu(\mathcal{P}^n(x)) \approx e^{-nh} \quad \text{combinatorial; needs a partition} \\ \text{Brin–Katok:} & \mu(B_n(x, \varepsilon)) \approx e^{-nh} \quad \text{geometric; needs a metric} \end{array}$$

Brin–Katok is the version we need, and it is worth saying why now rather than later: it speaks the language of derivatives. To apply it to a smooth system we need only understand *the shape of a Bowen ball*, and that is a question about Df . This is the hinge on which the second lecture turns.

5 End of Lecture 1

≈ 2 minutes

Summarize with a single display:

$$h_\mu(T) = \underbrace{\lim_n \frac{1}{n} H_\mu(\mathcal{P}_0^{n-1})}_{\text{information per unit time}} = \underbrace{\lim_n -\frac{1}{n} \log \mu(\mathcal{P}^n(x))}_{\text{SMB}} = \underbrace{\lim_{\varepsilon \rightarrow 0} \lim_n -\frac{1}{n} \log \mu(B_n(x, \varepsilon))}_{\text{Brin–Katok}}.$$

Three descriptions of one number: an average over the space, a limit along a single typical orbit, and a rate of shrinking of dynamical balls.

Then point out what is missing. Nothing so far has used a derivative. We computed $h_\mu = \log 2$ for the doubling map by counting binary digits, and separately noticed that $\log |T'| = \log 2$, with no explanation offered. Lecture 2 supplies the explanation.

Lecture 2: Where entropy comes from

6 Lyapunov exponents

≈ 14 minutes

6.1 One dimension: the chain rule plus Birkhoff

Let f be a C^1 map of the circle and μ an ergodic invariant measure. Ask how fast a small error grows:

$$\frac{1}{n} \log |(f^n)'(x)| \stackrel{\text{chain rule}}{=} \frac{1}{n} \log \prod_{i=0}^{n-1} |f'(f^i x)| = \frac{1}{n} \sum_{i=0}^{n-1} \log |f'(f^i x)|.$$

The right-hand side is a Birkhoff average of the fixed function $\log |f'|$, so the ergodic theorem gives, for μ -a.e. x ,

$$\lambda = \lim_{n \rightarrow \infty} \frac{1}{n} \log |(f^n)'(x)| = \int \log |f'| d\mu.$$

► *On the board:* this derivation. It is two lines and worth stating as a slogan: *the chain rule turns a product into a sum, and the ergodic theorem turns the sum into an integral.* That is the entire content of Lyapunov exponents in one dimension.

6.2 Two computations

For the doubling map $|T'| \equiv 2$, so $\lambda = \log 2$, and we computed $h_\mu(T) = \log 2$. Equal.

For the Gauss map $T(x) = \{1/x\}$ with its invariant density $\frac{1}{\log 2} \frac{1}{1+x}$, we have $|T'(x)| = 1/x^2$ and

$$\lambda = \int_0^1 \log \frac{1}{x^2} \frac{dx}{(1+x) \log 2} = \frac{2}{\log 2} \int_0^1 \frac{\log(1/x)}{1+x} dx = \frac{\pi^2}{6 \log 2} \approx 2.373.$$

This is a theorem about continued fractions. If p_n/q_n are the convergents of a typical x then $\frac{1}{n} \log q_n \rightarrow \pi^2/(12 \log 2) \approx 1.1866$, Lévy's constant, and the factor of two is because the n -th inverse branch of T has derivative of size q_n^{-2} . The entropy of the Gauss map is also $\pi^2/(6 \log 2)$. Equal again.

Two for two. The pattern is not a coincidence, but we do not yet have the tools to say why.

6.3 Higher dimensions

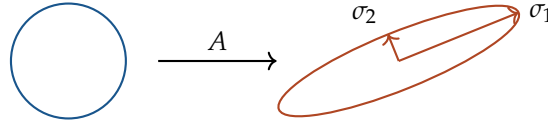
In dimension d the chain rule produces a product of *matrices*,

$$D_x f^n = D_{f^{n-1}x} f \cdots D_{fx} f \cdot D_x f,$$

which do not commute, so there is no “log of a product is a sum”. The right object is not a number but an ellipsoid: a linear map A carries the unit ball to an ellipsoid whose semi-axes are the *singular values*

$$\sigma_1(A) \geq \sigma_2(A) \geq \cdots \geq \sigma_d(A) \geq 0,$$

the square roots of the eigenvalues of $A^\top A$. These record the stretching in each of d orthogonal directions.



So the question becomes: does $\frac{1}{n} \log \sigma_i(D_x f^n)$ converge?

Theorem 6.1 (Oseledets 1968, multiplicative ergodic theorem). *Let f be a C^1 diffeomorphism of a compact manifold M^d and μ an ergodic invariant measure. There exist numbers $\lambda_1 > \dots > \lambda_r$ with multiplicities $m_1, \dots, m_r$ summing to d , and for μ -a.e. x a measurable splitting*

$$T_x M = E_1(x) \oplus \dots \oplus E_r(x), \quad \dim E_i = m_i, \quad D_x f E_i(x) = E_i(fx),$$

such that $\lim_{n \rightarrow \pm\infty} \frac{1}{n} \log \|D_x f^n v\| = \lambda_i$ for every $v \in E_i(x) \setminus \{0\}$. Moreover $\frac{1}{n} \log \sigma_i(D_x f^n) \rightarrow \lambda_i$, the exponents listed with multiplicity.

Do not attempt the proof. Describe it as an ergodic-theoretic singular value decomposition, and give the one picture that matters: an infinitesimal ball at x becomes, after n steps, an ellipsoid with semi-axes of size $e^{n\lambda_i}$.

6.4 Examples and a sanity check

For the cat map $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ the derivative is the same matrix everywhere, with eigenvalues $\frac{3 \pm \sqrt{5}}{2}$, so $\lambda_1 = \log \frac{3+\sqrt{5}}{2} \approx 0.9624$ and $\lambda_2 = -\lambda_1$. For a rotation $Df \equiv \text{Id}$ and $\lambda = 0$; compare $h = 0$.

Since $\prod_i \sigma_i(A) = |\det A|$, summing the exponents with multiplicity gives

$$\sum_i m_i \lambda_i = \int_M \log |\det Df| d\mu.$$

In particular if μ is volume and f preserves it then $\sum_i m_i \lambda_i = 0$: whatever is stretched in one direction is squeezed in another. This is a useful check and it explains why the cat map's exponents came out negatives of one another.

7 Ruelle's inequality

≈ 24 minutes

7.1 Statement

Theorem 7.1 (Ruelle 1978). *Let f be a C^1 diffeomorphism of a compact Riemannian manifold M^d and let μ be any f -invariant Borel probability measure. Then*

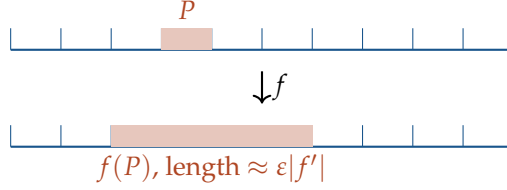
$$h_\mu(f) \leq \int_M \sum_i m_i \lambda_i^+(x) d\mu(x), \quad a^+ := \max(a, 0).$$

There is no hypothesis on μ beyond invariance. The right-hand side is the average rate of volume growth in the expanding directions.

The slogan to lead with, and to repeat at each step of the proof: **one cannot distinguish more orbits than the map creates room for**. Entropy counts how many orbit segments are separable; expansion is what separates them. The proof is that sentence made quantitative.

7.2 Warm-up: one dimension, in full

Let $f : \mathbb{T} \rightarrow \mathbb{T}$ be C^1 and partition $\mathbb{T}$ into about $1/\varepsilon$ intervals of length ε .



The image $f(P)$ is an interval of length at most $\varepsilon \sup_P |f'|$, so it meets at most

$$\frac{\varepsilon \sup_P |f'|}{\varepsilon} + 2 \leq 3 \max(\sup_P |f'|, 1)$$

cells of the partition. In other words, *the number of cells the image spreads across is the expansion factor*. Everything that follows is this count in d dimensions, together with the bookkeeping needed to convert $|f'|$ into λ .

7.3 Step 1: entropy is bounded by a counting number

Lemma 7.2. *Let T be an invertible measure-preserving map and $\mathcal{P}$ a finite partition. Then*

$$h_\mu(T, \mathcal{P}) \leq H_\mu(\mathcal{P} \mid T\mathcal{P}) \leq \sum_{Q \in \mathcal{P}} \mu(Q) \log N(Q), \quad N(Q) := \#\{P \in \mathcal{P} : P \cap TQ \neq \emptyset\}.$$

Proof. First, $h_\mu(T, \mathcal{P}) = h_\mu(T^{-1}, \mathcal{P})$: applying the measure-preserving map T^{n-1} carries $\bigvee_{i=0}^{n-1} T^{-i}\mathcal{P}$ to $\bigvee_{i=0}^{n-1} T^i\mathcal{P}$, so the two joins have the same entropy for every n , hence the same limit.

Second, for any measure-preserving S ,

$$h_\mu(S, \mathcal{P}) = \lim_n H_\mu\left(\mathcal{P} \mid \bigvee_{i=1}^n S^{-i}\mathcal{P}\right) \leq H_\mu(\mathcal{P} \mid S^{-1}\mathcal{P}),$$

since conditioning on more can only decrease entropy. Take $S = T^{-1}$, so that $S^{-1}\mathcal{P} = T\mathcal{P}$.

Third, decompose over the atoms of $T\mathcal{P}$, which are the sets TQ for $Q \in \mathcal{P}$:

$$H_\mu(\mathcal{P} \mid T\mathcal{P}) = \sum_Q \mu(TQ) H_\mu(\mathcal{P} \mid TQ).$$

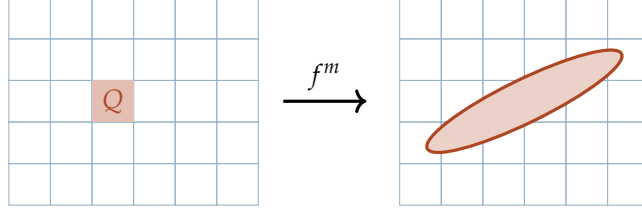
Conditioned on the atom TQ , the distribution of $\mathcal{P}$ is supported on at most $N(Q)$ cells, so by the crude bound (1) its entropy is at most $\log N(Q)$. Finally $\mu(TQ) = \mu(Q)$ by invariance and invertibility. $\square$

The orientation here matters and is easy to get backwards. Bounding $H_\mu(\mathcal{P} \mid f^{-1}\mathcal{P})$ instead would count how many cells $f^{-1}Q$ meets and would deliver the same theorem applied to f^{-1} — true, but stated in terms of the negative exponents, which reads backwards in a lecture. The version above counts how many cells the *image* $f(Q)$ spreads across, which is what the audience expects.

Note what has happened: all measure theory has been discharged. What remains is pure geometry.

7.4 Step 2: the geometry of the count

Fix $m \geq 1$ and apply Lemma 7.2 to $T = f^m$. Let $\mathcal{P}$ be a partition into roughly cubical cells of diameter at most ε .



Since f^m is C^1 , for ε small — depending on m , which is legitimate because we send $\varepsilon \rightarrow 0$ first — the image $f^m(Q)$ lies in an ellipsoid centred at $f^m(x)$ with semi-axes comparable to $\varepsilon \sigma_i(D_x f^m)$. An ellipsoid with semi-axes a_i meets at most $\prod_i (a_i/\varepsilon + 2)$ cells of diameter ε , whence

$$\log N(Q) \leq \sum_{i=1}^d \log^+ \sigma_i(D_x f^m) + C_d, \quad C_d = d \log 6,$$

for any $x \in Q$. The constant C_d depends only on the dimension — crucially, not on m and not on ε . That is the whole trick, as the next step shows.

7.5 Step 3: send $\varepsilon \rightarrow 0$, then $m \rightarrow \infty$

Combining Steps 1 and 2, and using that $x \mapsto \sum_i \log^+ \sigma_i(D_x f^m)$ is continuous and hence uniformly continuous on M , every partition of sufficiently small diameter satisfies

$$h_\mu(f^m, \mathcal{P}) \leq \int_M \sum_{i=1}^d \log^+ \sigma_i(D_x f^m) d\mu(x) + C_d.$$

Now invoke two standard facts: $h_\mu(f^m) = \lim_{\text{diam } \mathcal{P} \rightarrow 0} h_\mu(f^m, \mathcal{P})$, and $h_\mu(f^m) = m h_\mu(f)$. Therefore, for every m ,

$$h_\mu(f) \leq \int_M \frac{1}{m} \sum_{i=1}^d \log^+ \sigma_i(D_x f^m) d\mu(x) + \frac{C_d}{m}.$$

The additive constant is now divided by m and dies. This is what makes a crude covering bound sharp, and it is the step worth pausing on.

It remains to pass to the limit inside the integral. By Oseledets, $\frac{1}{m} \log \sigma_i(D_x f^m) \rightarrow \lambda_i(x)$ almost everywhere, so

$$\frac{1}{m} \log^+ \sigma_i(D_x f^m) = \max \left(\frac{1}{m} \log \sigma_i(D_x f^m), 0 \right) \longrightarrow \lambda_i^+(x).$$

For domination, set $L := \log^+ \max_{x \in M} \|D_x f\|$, finite by compactness and C^1 -ness; then

$$0 \leq \frac{1}{m} \log^+ \sigma_i(D_x f^m) \leq \frac{1}{m} \log^+ \|D_x f^m\| \leq L \quad \text{for all } m.$$

Dominated convergence applies, and letting $m \rightarrow \infty$ yields

$$h_\mu(f) \leq \int_M \sum_i m_i \lambda_i^+(x) d\mu(x). \quad \blacksquare$$

7.6 What the proof used, and where the slack is

It used only that conditioning reduces entropy and that $H \leq \log(\#\text{atoms})$ — the two crudest facts about Shannon entropy; that f^m maps small balls to near-ellipsoids; Oseledets, to identify the limit; and $h_\mu(f^m) = mh_\mu(f)$, to kill the dimensional constant. It used no hypothesis on μ beyond invariance, so the inequality holds for μ supported on a single periodic orbit or on a fractal attractor.

Where is the slack? In Step 1 we replaced a conditional distribution by the *uniform* distribution on the cells the image can reach, via $H \leq \log N$. Equality there demands that the measure spread evenly over all of those cells. That is precisely the condition Pesin's theorem will impose, and saying so here makes the next section feel inevitable rather than new.

8 Pesin's entropy formula

≈ 20 minutes

8.1 Statement

Theorem 8.1 (Pesin 1977). *Let f be a $C^{1+\alpha}$ diffeomorphism of a compact manifold and let μ be an f -invariant probability measure that is **absolutely continuous with respect to Lebesgue**. Then*

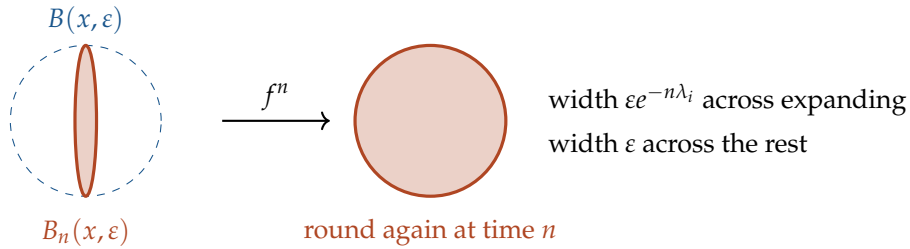
$$h_\mu(f) = \int_M \sum_i m_i \lambda_i^+(x) d\mu(x).$$

For a smooth system preserving a smooth measure, then, the rate at which information is created equals the total rate of expansion. Both hypotheses do real work: $C^{1+\alpha}$ rather than C^1 (there is a counterexample of Pugh), and absolute continuity, without which the statement is false, as we shall see in some detail.

8.2 The idea of the proof

By Brin–Katok, $h_\mu(f) = \lim_{\varepsilon \rightarrow 0} \lim_n -\frac{1}{n} \log \mu(B_n(x, \varepsilon))$, so we need only the *shape* of a Bowen ball.

To stay within ε of the orbit of x for n steps: in an expanding direction, with $\lambda_i > 0$, an initial displacement δ has grown to $\delta e^{n\lambda_i}$, so we need $\delta \lesssim \varepsilon e^{-n\lambda_i}$; whereas in a contracting or neutral direction, with $\lambda_i \leq 0$, displacements do not grow and any $\delta \leq \varepsilon$ will do. There is no constraint at all in those directions.



So $B_n(x, \varepsilon)$ is approximately an ellipsoid with semi-axes $\varepsilon e^{-n\lambda_i}$ in the expanding directions and ε in the others, and therefore

$$\text{Leb}(B_n(x, \varepsilon)) \approx \varepsilon^d \exp\left(-n \sum_i m_i \lambda_i^+\right).$$

Here is where absolute continuity enters. If $\mu \ll \text{Leb}$ with a density comparable to a constant on the relevant scales, then $\mu(B_n) \approx c \text{Leb}(B_n)$, so

$$-\frac{1}{n} \log \mu(B_n(x, \varepsilon)) \longrightarrow \sum_i m_i \lambda_i^+,$$

and by Brin–Katok the left side converges to $h_\mu(f)$. That is the formula.

A real proof must control the density along orbits, which is where $C^{1+\alpha}$ and the absolute continuity of the stable foliation are needed. But the heuristic is honest: this really is why the theorem is true, and it is why Brin–Katok was worth proving in Lecture 1.

8.3 Why absolute continuity is not a technicality

The computation compared $\mu(B_n)$ with $\text{volume}(B_n)$. If μ is singular it may sit on a fractal subset of the ellipsoid, in which case $\mu(B_n)$ decays more slowly than the volume and Ruelle’s inequality is strict. Two examples on the doubling map, where $\lambda \equiv \log 2$ always.

A point mass. Take $\mu = \delta_0$, invariant since $T(0) = 0$. Then $h_\mu(T) = 0$ while $\lambda = \log 2$. A fixed point produces no information though the map expands as fast as ever: maximal failure.

Bernoulli measures. Identify $\mathbb{R}/\mathbb{Z}$ with $\{0, 1\}^{\mathbb{N}}$ by binary expansion, so that T becomes the shift, and let μ_p be the Bernoulli($p, 1-p$) measure. By the law of large numbers μ_p -a.e. point has digit frequency p , so for $p \neq \frac{1}{2}$ these measures are mutually singular and singular with respect to Lebesgue $= \mu_{1/2}$. And, from Lecture 1,

$$h_\mu(T) = H(p) < \log 2 = \lambda \quad (p \neq \tfrac{1}{2}).$$

This second example is the one to spend time on, because the gap is not merely an inequality — it is a dimension:

$$\dim_H \mu_p = \frac{H(p)}{\log 2} = \frac{h}{\lambda}.$$

The measure fills only an h/λ fraction of the available directions, and Pesin’s equality holds exactly when that fraction is 1.

8.4 The whole story in one table

system	measure	h_μ	$\sum m_i \lambda_i^+$	equality?
rotation R_α	Lebesgue	0	0	yes
doubling $x \mapsto 2x$	Lebesgue	$\log 2$	$\log 2$	yes
doubling $x \mapsto 2x$	Bernoulli(p)	$H(p)$	$\log 2$	no
doubling $x \mapsto 2x$	δ_0	0	$\log 2$	no
Gauss map	Gauss measure	$\pi^2/6 \log 2$	$\pi^2/6 \log 2$	yes
cat map	Lebesgue	$\log \frac{3+\sqrt{5}}{2}$	$\log \frac{3+\sqrt{5}}{2}$	yes
cat map	periodic orbit	0	$\log \frac{3+\sqrt{5}}{2}$	no
solenoid	SRB measure	$\log 2$	$\log 2$	yes
Hénon (Benedicks–Young)	SRB measure	> 0	$= h_\mu$	yes

The rows marked “yes” are exactly those where the measure is smooth along the expanding directions. That observation is a theorem.

8.5 What equality really means: SRB measures

Theorem 8.2 (Ledrappier–Strelcyn; Ledrappier–Young, 1984–85). *For $f \in \text{Diff}^{1+\alpha}(M)$ and μ invariant with a positive exponent, Pesin’s formula holds if and only if μ has absolutely continuous conditional measures on unstable manifolds.*

Such measures are called *SRB*, after Sinai, Ruelle and Bowen. They need not be absolutely continuous on M — an SRB measure on the Hénon attractor is singular, living on a set of dimension less than two — but they are smooth *in the expanding directions*, which is all the Bowen ball argument ever needed.

SRB measures are also exactly the physically observable ones: for a set of initial conditions of positive Lebesgue measure, time averages converge to the SRB average. So Pesin’s formula characterizes the measures one would actually see in an experiment, which is a good note to end the theory on.

8.6 The general picture: Ledrappier–Young

For a general invariant measure the deficit in Ruelle’s inequality is measured by dimension.

Theorem 8.3 (Ledrappier–Young 1985). *For $f \in \text{Diff}^{1+\alpha}(M)$ and μ ergodic,*

$$h_\mu(f) = \sum_{i:\lambda_i>0} \lambda_i \gamma_i,$$

where $\gamma_i \in [0, m_i]$ is the dimension of μ transverse to the i -th expanding direction.

Taking $\gamma_i = m_i$ for all i recovers Pesin’s formula: the measure is full-dimensional along unstables. In our running example there is one exponent $\lambda = \log 2$ and $h = \lambda \gamma$ with $\gamma = \dim_H \mu_p = H(p) / \log 2$. The identity to write on the board is

$$\text{entropy} = \text{exponent} \times \text{dimension},$$

which in various forms organizes much of the modern subject.

8.7 Back to the beginning

Close by returning to $T(x) = 2x$ on $\mathbb{R}/\mathbb{Z}$ with Lebesgue measure.

It destroys one bit per step: that is $h_\mu(T) = \log 2$, which is simultaneously the Shannon entropy of the digit process, the Kolmogorov–Sinai entropy by the generator theorem, and the decay rate of Bowen balls by Brin–Katok. It expands lengths by a factor of two: that is $\lambda = \int \log |T'| = \log 2$. The two agree because Lebesgue measure is smooth, so a Bowen ball — an interval of length $\varepsilon 2^{-n}$ — carries measure proportional to its length; that is Pesin’s formula. And they would not agree for the singular measures μ_p , where instead $h = \lambda \cdot \dim_H \mu_p$.

Chaos manufactures information at exactly the rate it stretches space — provided the measure is willing to spread out.

9 Asides, if a section runs short

Why SMB is true

The Bernoulli case was the law of large numbers applied to $-\log p_{x_0}$. In general the terms are dependent, but the idea survives. Writing $I_n(x) = -\log \mu(\mathcal{P}^n(x))$, one telescopes

$$\frac{1}{n}I_n = \frac{1}{n} \sum_{k=0}^{n-1} g_k(T^k x),$$

where g_k is the information in $\mathcal{P}$ given $\mathcal{P}_1^k$. Martingale convergence gives $g_k \rightarrow g_\infty$ almost everywhere and in L^1 , with g_∞ the information in $\mathcal{P}$ given the entire past; Birkhoff gives $\frac{1}{n} \sum_{k=0}^{n-1} g_\infty(T^k x) \rightarrow \int g_\infty d\mu = h_\mu(T, \mathcal{P})$; and a maximal inequality, showing $\sup_k g_k \in L^1$, controls the difference. So: the ergodic theorem plus the martingale convergence theorem, with the work in the gluing.

Entropy as algorithmic complexity

Let $K(w)$ be the *Kolmogorov complexity* of a finite word w , the length in bits of the shortest program that outputs w . This has nothing to do with probability; it is a property of the individual string.

Theorem 9.1 (Brudno 1983). *For an ergodic measure μ on a symbolic system and μ -a.e. x ,*

$$\lim_{n \rightarrow \infty} \frac{K(x_1 x_2 \cdots x_n)}{n} = h_\mu(\sigma) \quad (\text{in bits}).$$

So entropy is also the compressibility of a typical orbit, giving a third pointwise description alongside SMB and Brin–Katok. The same phenomenon recurs for Hausdorff dimension: the point-to-set principle of J. Lutz and N. Lutz identifies the dimension of a set with the complexity rate of its individual points.

Entropy as a growth rate of wealth

Betting on X at fair odds and reinvesting everything each round, the optimal strategy is to bet proportionally, $b_i = p_i$, and wealth then grows like $W_n \approx e^{n(\log k - H(p))}$. Entropy is exactly the shortfall from the maximum possible growth rate: information you have that the market does not is worth money, at an exchange rate of one doubling of capital per bit (Kelly, 1956).

References

Books. Cover & Thomas, *Elements of Information Theory*, for §2. Walters, *An Introduction to Ergodic Theory*, for §3. Mañé, *Ergodic Theory and Differentiable Dynamics*, which includes Ruelle’s inequality. Barreira & Pesin, *Nonuniform Hyperbolicity*, for Oseledets and Pesin theory. Katok & Hasselblatt, *Introduction to the Modern Theory of Dynamical Systems*.

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83–87. F. Ledrappier and L.-S. Young, *The metric entropy of diffeomorphisms I, II*, Ann. of Math. 122 (1985). L.-S. Young, *What are SRB measures, and which dynamical systems have them?*, J. Stat. Phys. 108 (2002), 733–754 — an outstanding survey, and the right place to start.